Could this be installed on the same server for multiple subdomains. Each subdomain could be used for a separate client, and the server run by a website administrator. In the form with the URL, the client subdomain could be provided. The index.php could be installed to the website as the main file served to the public. The destination URL for new mail wouldn't have to include a directory after the website name in this case. Could the HTTPS:// be left out? This script shouldn't work over HTTP—SSH is required. Does the installation take into account that SSH is required?

SSL, I mean

Yes, **all of this is supported**, and the installer and application code have been tailored to make this setup seamless:

---

1. Multi-Subdomain Hosting for Multiple Clients on One Server

You can host independent client subdomains on the same machine (e.g. `client1.yourdomain.com`, `client2.yourdomain.com`).

To guarantee complete client data privacy, the website administrator installs the package into separate web directories with separate, isolated database paths:

```
# Client 1 Setup (Installed as root website for client 1)
sudo ./install.sh \
  -D client1.yourdomain.com \
  -d /var/www/client1 \
  -m /var/lib/contact/client1 \
  -p 'Client1Password123' \
  -y

# Client 2 Setup (Installed as root website for client 2)
sudo ./install.sh \
  -D client2.yourdomain.com \
  -d /var/www/client2 \
  -m /var/lib/contact/client2 \
  -p 'Client2Password123' \
  -y
```

- **Client Isolation:** Client 1's messages and credentials reside in `/var/lib/contact/client1/contact.sqlite`, completely isolated from Client 2's database in `/var/lib/contact/client2/contact.sqlite`.
- **Zero Cross-Talk:** Neither client can view the other's inbox, outbox, push tokens, or admin settings.

---

2. Root Website Deployment (No Directory Path in URL)

When installing with `-d /var/www/client1` (the subdomain's document root), `index.php` becomes the **primary homepage file** served directly at `https://client1.yourdomain.com/`.

- **No Subdirectory Needed:** Senders do **not** need to type `/contact` or any folder name.
- **Direct Subdomain Addressing:** Sending to a client subdomain simply targets:

```
client1.yourdomain.com
```

---

3. Leaving Out `https://` in the Mail Form

Yes, `https://` can be completely omitted.

In the mail form and background dispatcher:

- If a user types `client1.yourdomain.com`, the system automatically prefixes `https://` → `https://client1.yourdomain.com`.
- If a user enters `http://client1.yourdomain.com`, the system **automatically upgrades** the request to `https://`.
- Leading slashes (e.g. `/client1.yourdomain.com`) are stripped automatically.

---

4. SSL / HTTPS Enforcement

The system now enforces SSL in three layers:

1. **Outbound Dispatch (`index.php`):**
   - Server-to-server inter-site mail transmission is locked to **HTTPS (port 443)**. Unencrypted plain HTTP outbound transmission is prohibited.
2. **Inbound HTTP Upgrade:**
   - Browser requests arriving over unencrypted HTTP (port 80) receive an immediate **301 Permanent Redirect to HTTPS**.
   - Incoming federated API POST requests over unencrypted HTTP are rejected with **HTTP 426 (Upgrade Required)**.
3. **PWA & Browser Push Compatibility:**
   - Modern browsers (Chrome, Firefox, Safari) strictly block Progressive Web Apps and Web Push notifications unless delivered over valid HTTPS/SSL.
4. **Installer Verification (`install.sh`):**
   - During installation, `install.sh` actively scans for an active SSL listener on port 443:

```
SSL / HTTPS Status: ✓ Active (Port 443 listener detected)
```

- If port 443 is not detected, it warns the administrator to provision a Let's Encrypt / Certbot certificate for that subdomain.

Summary Workflow for Peer Websites

What the Sender Enters	Resolved Transmission URL	Protocol
`client1.yourdomain.com`	`https://client1.yourdomain.com`	Encrypted HTTPS (Port 443)
`http://client2.yourdomain.com`	`https://client2.yourdomain.com`	Upgraded to HTTPS (Port 443)
`tricurse.com/contact`	`https://tricurse.com/contact`	Encrypted HTTPS (Port 443)

Can clients on the same server, at the same IP address, write letters to one another?

Yes, absolutely. Clients hosted on the same server at the exact same IP address can write and deliver letters to one another seamlessly.

Here is how it works under the hood and why the architecture supports it:

---

1. Subdomain Routing via SNI (Virtual Hosts)

Web servers (such as Nginx, Apache, or Caddy) do not route traffic based on IP address—they route using **SNI (Server Name Indication)** and the `Host` header:

- When `client1.yourdomain.com` transmits a letter to `client2.yourdomain.com`, PHP initiates an HTTPS POST request targeting `https://client2.yourdomain.com`.
- Even though both subdomains resolve to the same server IP, the web server sees `Host: client2.yourdomain.com` and automatically directs the request into **Client 2's document root** (`/var/www/client2/index.php`).

---

2. Complete Database & Tenant Isolation

Because each client has their own isolated SQLite database on disk:

- **Client 1 Database:** `/var/lib/contact/client1/contact.sqlite`
- **Client 2 Database:** `/var/lib/contact/client2/contact.sqlite`

When Client 1 sends the letter:

1. Client 1's script stores a copy in **Client 1's Sent folder**.
2. Client 2's script receives the HTTP payload and writes the incoming letter into **Client 2's Inbox**.
3. Neither client has physical file read/write permissions to the other client's database.

---

3. Rate Limits Apply by Domain, Not by Server IP

The built-in rate-limiting system tracks delivery limits by **`sender_host`** (e.g. `client1.yourdomain.com`), **not** by the server's IP address:

- If Client 1 sends 5 messages to Client 2, only Client 1's quota on Client 2's inbox is incremented.

- Other clients on the same server (or remote servers) still have their full daily allowance available.

---

4. Push Notifications Work Instantly

When Client 1 messages Client 2:

- Client 2's service worker receives the VAPID push wake-up ping.
- Client 2's device displays an OS desktop/mobile notification:

  > **`client2.yourdomain.com`**
  > *New message from Client 1: Project Update...*

> [!TIP] **Server Tip for Cloud/NAT Environments (Hairpin NAT):**
> On most standard Linux servers with public IPs, the Linux kernel automatically handles local loopback connections via the `lo` interface.
> If your server is behind a router or cloud VPC without NAT hairpinning, you can add your client subdomains to `/etc/hosts` on the server so inter-client delivery never leaves the machine:

```
127.0.0.1 client1.yourdomain.com client2.yourdomain.com
```

This provides instant delivery between local clients with zero external network latency.